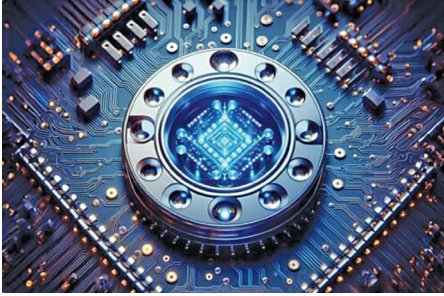


8-photon chip marks quantum leap

The Electronics and Telecommunications Research Institute (ETRI) in South Korea has developed a photonic quantum circuit chip capable of controlling up to eight photons. This chip advances multipartite entanglement research,



Researchers have created a quantum circuit chip that controls photons for advanced quantum computing, achieving up to 6-qubit entanglement (Credit: SciTechDaily.com)

which is crucial for quantum computing. It used silicon-photon quantum circuits to demonstrate 6-qubit entanglement on this new 8-photon chip, progressing from earlier 2-qubit and 4-qubit

experiments. The chip integrates eight photon sources and ~40 optical switches for photon path control, enabling room-temperature operations with energy efficiency. Experimental successes include observing the Hong-Ou-Mandel effect. Future work focuses on scaling up to 32 qubits and enabling cloud-based quantum services.

Hydrovoltaic cells can generate electricity without sunlight

Engineers at the Chinese Academy of Sciences have unveiled a hydrovoltaic cell, the Hermetic hydrovoltaic cell (HHC), capable of generating electricity without sunlight or substantial water. Unlike conventional hydrovoltaic cells reliant on water-surface interactions, the HHC utilises ambient heat as its energy source, overcoming the limitations of sunlight dependency. Encased in a hermetically sealed structure, it employs a dual-layered carbon black and tissue paper wicking mechanism, maintaining operation through natural capillary action and thermal water cycling. It provides up to 160 hours of power with minimal water. Sunlight exposure further boosts efficiency by enhancing moisture gradients. Durable and cost-effective, the HHC is ideal for arid regions, disaster relief, and off-grid applications, advancing renewable energy sustainability.

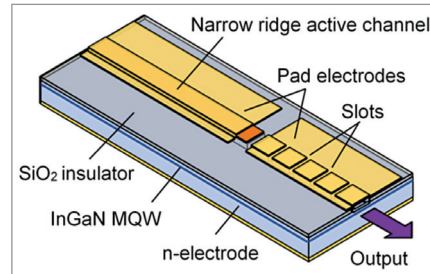
Optical biosensor diagnoses monkeypox

Researchers at UC San Diego School of Medicine, in collaboration with Boston University, have developed an optical biosensor for rapid monkeypox detection, advancing

diagnostic technology beyond traditional lab setups. It leverages pixel-diversity interferometric reflectance imaging sensor (PD-IRIS) technology. It utilises silicon chips coated with monoclonal antibodies to bind viral proteins in patient samples. Light exposure creates interference patterns, allowing precise virus identification. In under 20 minutes, including sample processing, monkeypox is distinguished from viruses similar to herpes simplex. Scalable for broader viral diagnostics, this innovation addresses global outbreak challenges.

Adjustable wavelength disinfection laser

Scientists at Osaka University have developed a compact, tunable-wavelength blue semiconductor laser, advancing far-ultraviolet (UV) light technology. This laser, operating in the 405nm band and modifiable for 460nm, leverages



InGaN tuneable single-mode laser with periodically slotted structure (Credit: Taisei Kusui)

a periodically slotted nitride semiconductor structure. Unlike bulky ultrashort-pulse lasers, this is compact, cost-effective, and ideal for next-gen disinfection technologies. It supports high-

efficiency wavelength conversion to far-UV (below 230nm) using nonlinear crystals like aluminium nitride and SrB4O7 in microcavities. Due to size and cost, previous systems needed more practicality, but this offers versatility, compactness, and long-term reliability for appliance applications, enhancing public health with cleaner environments.

Solar balloon power generator

Researchers from Mälardalen University, Southwest Jiaotong University, and Guizhou University have developed the balloon-integrated photovoltaic system (BIPVS) for decentralised solar energy generation. This hybrid balloon uses helium and air for lift. Its transparent top focuses sunlight via refractive properties, while photovoltaic cells beneath convert it into electricity. Protected from weather, the system transmits power via an electrical cord and stabilises using cables. An exhaust valve regulates gas pressure, ensuring stability. Testing shows one balloon generates 3.5-4GWh/month, offering scalable, efficient energy production, even in challenging climates, addressing traditional solar limitations like snow and poor sunlight.